Tight Clouds versus Locally Finite Divisors
An Exact Obstruction to Direct Rank-Growing Root Limits

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July 2026

Abstract

Global centered-RMS normalization makes every empirical root measure have unit second moment, and RH-224 therefore proves uniform tightness. This paper shows why that positive probability result cannot be promoted directly to a holomorphic zero-divisor limit.

Let Λ_n be finite clouds with ranks $d_n \to \infty$, let $\nu_n = \sum_{\lambda\in\Lambda_n} \delta_\lambda$ be their integer root divisors, and let $\mu_n = d_n^{-1}\nu_n$ be their empirical probability measures. If $\{\mu_n\}$ is uniformly tight, one fixed compact set contains at least half of every cloud. Hence ν_n has mass at least $d_n/2$ on a fixed compact set, which diverges. The divisors cannot converge vaguely to a locally finite divisor, and they cannot be zero divisors of a locally uniformly convergent nonzero holomorphic family.

In the RH-222 atlas all normalized roots lie in $|z| \le 2$, while all raw Hardy-scaled resonances lie in $|z| < 1$ with maximum modulus 0.86860. Compact divisor counts grow from 4 to 35 on the left and from 4 to 34 on the right.

The obstruction rejects the resonances themselves, normalized or raw, as the zeros of the sought growing determinant. It does not reject the reciprocal Fredholm zeros $1/\lambda$, which can escape to infinity when resonances tend to zero. This distinction redirects the next layer to the correct spectral variable.

1 Two measures carried by the same roots

For a finite multiset

$$\Lambda_n = \{\lambda_{n,1}, \dots, \lambda_{n,d_n}\} \subset \mathbb{C},$$

there are two natural measures:

$$\nu_n = \sum_{j=1}^{d_n} \delta_{\lambda_{n,j}}, \qquad \mu_n = \frac{1}{d_n}\nu_n. \tag{1}$$

The first is the integer divisor measure. The second is the empirical probability measure.

Weak compactness of μ_n is a macroscopic statement in which each root has mass $1/d_n$. Local finiteness of ν_n is a microscopic statement in which each simple root has mass one. RH-224 proves tightness only for the first normalization [3].

2 The tightness obstruction

Theorem 2.1 (Tight rank growth is not a locally finite divisor)**.** *Suppose $d_n \to \infty$ and the empirical measures $\{\mu_n\}$ in (1) are uniformly tight. Then there is a compact set $K \subset \mathbb{C}$ such that*

$$\nu_n(K) \ge \frac{d_n}{2} \tag{2}$$

for every n. Consequently no subsequence of ν_n converges vaguely to a locally finite Radon measure.

Proof. Uniform tightness with $\varepsilon = 1/2$ supplies compact K such that $\mu_n(K) \geq 1/2$ for all n. Multiplying by d_n proves (2).

Choose a compactly supported continuous function $\phi \geq 0$ that equals one on K. Then

$$\int \phi \, d\nu_n \geq \nu_n(K) \longrightarrow \infty.$$

Vague convergence to a locally finite Radon measure ν would instead give the finite limit $\int \phi \, d\nu$. This is impossible. ☐

The fraction 1/2 is arbitrary. For any fixed $0 < \varepsilon < 1$, one compact set carries at least $(1 - \varepsilon)d_n$ divisor mass.

Corollary 2.2 (No direct nonzero holomorphic limit)**.** *Under the assumptions of Theorem 2.1, the Λ_n cannot be the complete zero divisors on $\mathbb{C}$ of holomorphic functions f_n converging locally uniformly to a nonzero entire function f, with zeros counted by algebraic multiplicity.*

Proof. On a compact neighborhood whose boundary avoids the zeros of f, local uniform convergence and Rouche's theorem make the number of zeros eventually equal to that of f; see Conway [1]. Equivalently, zero divisors of a nonzero locally uniform limit are locally bounded. This contradicts Theorem 2.1. ☐

The only holomorphic escape is degeneration to the identically zero function, which is not a spectral determinant.

3 Application to globally normalized clouds

RH-224 gives

$$\mu_n(|z| > R) \leq R^{-2}.$$

At $R = 2$, every normalized cloud has at least 75% of its roots in one fixed disk. Thus any all-level extension with ranks tending to infinity would satisfy

$$\nu_n(\overline{D(0,2)}) \geq \frac{3}{4} d_n \to \infty.$$

The exact theorem already rejects the direct normalized-root divisor. No numerical convergence assumption is needed.

This conclusion complements the fixed-quartic obstruction RH-219 [2]. RH-219 rules out degree four and repeated fixed support. Theorem 2.1 rules out a different failure mode: genuinely new roots are added, but a positive fraction remains in one compact region.

4 Finite physical audit

The sixteen-level clouds have strictly increasing ranks. The normalized and raw compact counts are:

channel	first rank	final rank	compact-count growth
left, normalized $\lvert z\rvert \leq 2$	4	35	31
right, normalized $\lvert z\rvert \leq 2$	4	34	30
left, raw $\lvert z\rvert \leq 1$	4	35	31
right, raw $\lvert z\rvert \leq 1$	4	34	30

Every audited normalized root lies in $|z| \le 2$. Every raw selected resonance lies in the unit disk, and the largest raw modulus is 0.868592.

The raw statement is finite. Although Markov resonances before Hardy scaling lie in the unit disk at fixed positive noise, the chosen 0.85 scaling could in principle move some bulk roots beyond one at smaller unobserved scales. The theorem-level direct-normalization obstruction does not depend on that finite raw-radius observation.

5 Why reciprocal zeros are different

For a compact operator with nonzero eigenvalues $\lambda_j \to 0$, the Fredholm determinant has zeros at

$$z_j = \lambda_j^{-1}. \tag{3}$$

The newly added reciprocal zeros can escape every compact set. Their integer divisor may therefore remain locally finite even though the resonance empirical measure is tight or even converges to a point at zero.

Inversion is singular at the resonance accumulation point. It does not preserve probability tightness, and Theorem 2.1 cannot be applied to the reciprocal cloud without a separate hypothesis.

This is precisely the fixed-noise architecture of the regularized determinant constructed in RH-7 [4]. The next paper restores the finite identity between the characteristic polynomial in λ and the Fredholm polynomial in z.

6 Route consequence

The following routes are now separated:

1. raw or centered resonances used directly as zeros: rejected;
2. repeated fixed factors: rejected by RH-219;
3. reciprocal resonance zeros at fixed noise: operator-theoretically valid;
4. a reciprocal small-noise divisor: still open and must pass local-count and omitted-tail tests.

No small-noise obstruction for reciprocal zeros is proved here. Gate A remains open, as do all self-adjoint, counting-law, arithmetic, and zeta identification stages.

References

[1] John B. Conway. *Functions of One Complex Variable.* Springer, 1978.

[2] Liang Wang. Why a fixed quartic cannot supply a growing spectral count. 2026. RH-219 preprint.

[3] Liang Wang. Global cloud gauge and empirical-root tightness. 2026. RH-224 preprint.

[4] Liang Wang. Irreversible gaussian cycle spectrum. 2026. RH-7 preprint.